

GE Healthcare Technical Bulletin



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Applicable to ISO 13485 QMS

TECHNICAL BULLETIN

Technical Bulletin Unique Identifier	SN DOC2835137
Title	Service Note for MR29_R04 software release
Subject	The MR29.1_R04 software was released to introduce software enhancements and fix existing MR29 software errors, the SPRs are summarized in Appendix A
Applicability (e.g., affected product, process)	For Product used 1.5T list as below: • SIGNA™ Artist configurations • SIGNA™ Voyager configurations • SIGNA™ Creator / Explorer configurations • SIGNA™ MR355 / MR360 configurations • Optima™ MR450W 1.5T configurations (includes 450W GEM and 450W Non-GEM configs) • 1.5T SIGNA™ HDxt (16 channel) configurations For Product used 3.0T list as below: • SIGNA™ Architect configurations • SIGNA™ Pioneer configurations • SIGNA™ Premier configurations • Discovery MR750/750w configurations
Date	08-May-2023
Originating Business (Select one)	☐ GEHC Digital (Imaging and Care Area Solutions, Value Based Care) ☐ Clinical Care Solutions (Ultrasound, Lifecare Solutions) ☒ Imaging (Magnetic Resonance, Molecular Imaging and Computed Tomography, X-ray, Interventional, Surgery, Womens Health, Global Services) ☐ Other: _____
Originating Modality	MR
Details	This service note informs the field that the MR29.1_R04 software has been released for above MR systems.
Tracking Number	ECR2357371; ECR2357372; ECR2357369; ECR2357370; ECR2361074; ECR2359767; ECR2357839; ECR2358881; ECR2358883; ECR2358884
Affected Service Publications	The A-Port 8ch TR knee coil is phased in the MR29.1_R04 release. The impacted service manual has been updated and will release along with incoming 1.5T MR30.1 software release. No 3.0T service documentation is affected.
Resolution	N/A
Technical Bulletin POC	Hou Jiannan (SSO:223053253)

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Effectivity

For Product used 1.5T list as below:

- SIGNA™ Artist configurations
- SIGNA™ Voyager configurations
- SIGNA™ Creator / Explorer configurations
- SIGNA™ MR355 / MR360 configurations
- Optima™ MR450W 1.5T configurations (includes 450W GEM and 450W Non-GEM configs)
- 1.5T SIGNA™ HDxt (16 channel) configurations

For Product used 3.0T list as below:

- SIGNA™ Architect configurations
- SIGNA™ Pioneer configurations
- SIGNA™ Premier configurations
- Discovery MR750/750w configurations

Details

This software is being introduced into forward production for systems shipping with MR29.1 software. Some of the software fixes include:

HCSDM00728572	System cannot detect HEC via cable p/n 5729973 changed under SCR-0131764
HCSDM00720680	[MR29.1_R04] [swtools]VX29.1_R02 (and RX29.1) Root Partition Full issue

The full list of software fixes is contained in **Appendix A** below.

Note: This Service note only contains the new changes related to service procedure introduced by the release of MR29.1_R04 software. For more common information and features with existing MR29.1 software, please refer to the 1.5T and 3.0T MR29.1 Iserv documents, DOC2520364 and DOC2536251.

FMI's

A mandatory business FMI67943 will be released for sites operating on software MR29.1_R01 and R02.

FMI 67943A provides software **VX29.1_R04** to 1.5T SIGNA Voyager sites with **VX29.1**.

FMI 67943B provides software **SV29.1_R04** to 1.5T SIGNA Creator/ Explorer sites with **SV29.1**.

FMI 67943C provides software **HD29.1_R04** to 1.5T SIGNA HDxt sites with **HD29.1**.

FMI 67943F provides software **DV29.1_R04** to 1.5T SIGNA Artist/Optima MR450w & 3.0T SIGNA Architect/Discovery MR750w/MR750 sites with **DV29.1**.

FMI 67943G provides software **PX29.1_R04** to 3.0T SIGNA Pioneer sites with **PX29.1 R01/R02**.

FMI 67943H provides software **RX29.1_R04** to 3.0T SIGNA Premier sites with **RX29.1**.

Note: On MR29.1_R04, there is a P file format change to fix a spectroscopy issue. However, this change missed to evaluate the impact on the MR guided focused Ultrasound function(FUS), so there is a temporary loss of this functionality on MR29.1_R04 SW. IB customers shall have the flexibility to refuse the FMI if they are using FUS.

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Option keys:

Four new options listed below were introduced in the MR29.1_R04 software release. These options will be updated in Service Manuals and Product Configuration Masters corresponding to their applicable products.

Note: The Air Recon DL option key was split into two option keys, airrcndl2d (original), and ARDLdwepi (new). For sites that already have M7009SA, if they have the Air Recon Option key, that this option key will need to be reloaded with the R04 software otherwise the customer will lose functionality.

Clinical Name	Mcat	Option Key Included
2D AIR Recon DL for China without DW-EPI	M7009SY	airrcndl2d [new]
2D AIR Recon DL for China with DW-EPI	M7009SX	airrcndl2d, ARDLdwepi [new]
2D AIR Recon DL DW-EPI	M7013SA	ARDLdwepi [new]
Breast Image Orientation	M7030AA	BreastImgO [new]
AIR Recon DL	M7009SA	airrcndl2d, ARDLdwepi [updated]

Coil:

In MR29.1_R04 software release, the 1.5T A-port TR knee coil was phased in Creator/Explorer/MR360 products, which will be setup as a new FRU in the incoming MR30.1 software release.

<i>New FRU Part Number</i>	<i>Description</i>
5932028-2	1.5T 8ch TR knee Array

When setting up this coil for MCQA test it should connect to the A port instead of P port. Please pay attention to its difference with P-port type when you refer to the product service method coil vendor manual section for detailed MCQA or replacement procedure. The update will be included with the 1.5T MR30.1 release.

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Rev	Page	Description of Content Changed	Reason for Change or Change Control Number	Author Name (SSO)
1	All	Initial Release	Initial Release	Hou Jiannan (223053253)
2	p2	Add note below the FMI part	MR29.1_R04 doesn't support FUS	Hou Jiannan (223053253)

Appendix A:

SPR#	Headline
HCSDM00712391	[MR29.1_R04] [PSD]AutoVoice failed for Disco 1st phase with SmartPrep
HCSDM00712392	[MR29.1_R04] [PSD]MAVRIC STIR minseqrfamp error with weight >160kg
HCSDM00712393	[MR29.1_R04] [PSD]xTG failed issue for newborn baby
HCSDM00712400	[MR29.1_R04] [PSD]Disable DISCO run_eval_from_get_flag for all 1.5T systems
HCSDM00712402	[MR29.1_R04] [PSD]DW PROP DUO shading on Explorer
HCSDM00712404	[MR29.1_R04] [PSD]3D FSPGR FS CV34 logic issue
HCSDM00712501	[MR29.1_R04] [3T] Vibrant sp sat scan time fix
HCSDM00712561	[MR29.1_R04] [Host]Operator can't record customer voice.
HCSDM00712563	[MR29.1_R04] [Host]Nothing is displayed in the Intensity Filter selection box after SSFSE setup
HCSDM00712564	[MR29.1_R04] [swtools] [Site issue] setNavShimSupply crash when using NAV shim PS
HCSDM00712641	MP-RAGE PROMO image shift after the previous MP-RAGE PROMO scan aborted
HCSDM00712643	Overrange on MP-RAGE brain protocol
HCSDM00712644	PSD timeout when adding SSRF imaging option to Ax VIBRANT high spatial resolution GE protocol
HCSDM00712645	3dtof gradient trip due to missing ISI on ICE
HCSDM00712668	[MR29.1_R04] [PSD] CUBE fineline reduction
HCSDM00712670	[MR29.1_R04] [PSD] HCSDM00707469: Slice ASSET for sagittal Vibrant has image stretch/shrink
HCSDM00712671	[MR29.1_R04] For FSE, Dynamic TG(CV38) & Blurring Cancellation (CV7) on at same time give noisy images
HCSDM00712673	[MR29.1_R04] 3dtof doesn't show MT effect
HCSDM00712678	Adding error message arguments in verify_scan_info.cpp to dump log information to psd trace
HCSDM00712682	[MR29.1_R04] [PSD] SSFSE abd T2 shading mitigation with PURE tuning for 3T
HCSDM00712688	Regular 3D SPGR/GRE show wrong slice resolution on the UI
HCSDM00712759	[SW.Lx.29.1_R04] [Recon] Update rdb_hdr_raw_pass_size in pfile header when CoilSwitch is ran in prescan
HCSDM00712798	[MR29.1_R04] [PSD] 3dradial ZTE scan failed with Resp. Trigger

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HCSDM00712800	[MR29.1_R04] [PSD] Prescan Cal PSD Encoding Center Shift
HCSDM00712966	[MR29.1_R04] [host] IB: Frequency Direction Saved in the Protocol is getting swapped in the scanning Session.
HCSDM00712967	[MR29.1_R04] [host] Exsi Crash and Scan crash in SetCollectPutCvs
HCSDM00712968	[MR29.1_R04] [host] ZTE protocol shows up advisory message for slice thickness during protocol load
HCSDM00713057	[MR29.1_R04] [PSD] Intleave needs to be disabled in Spacing UI for MUSE
HCSDM00713058	[MR29.1_R04] [PSD] C045 EPI protocol download failed
HCSDM00713060	[MR29.1_R04] [PSD] cfx/y/zfs are not 3.21 when ART is ON with DW-EPI with XRMW
HCSDM00713061	[MR29.1_R04] [PSD] epi2 timeout during HAST
HCSDM00713063	[MR29.1_R04] [PSD] type-in PSD epialtoff and epiRTaltoff fail to set pepolar =0 for hyperband scan
HCSDM00713064	[MR29.1_R04] [PSD] dacq_adjust update for DW-EPI with Hyperband
HCSDM00713065	[MR29.1_R04] [PSD] Severe A/P ghosting with Asset on GRE-EPI
HCSDM00713066	[MR29.1_R04] [PSD] Hyperband fMRI with spsp fatsat (opuser4) and thin slices
HCSDM00713082	[MR29.1_R04] [host]banding artifact worsened at site after 29.0 to 29.1 update. same issue in MR30.0
HCSDM00713083	[MR29.1_R04] [host] CS0 00833656: AIRx need extra option for Slice Orientation to meet with site's various slice order custom
HCSDM00713350	[SW.Lx.29.1_R04] [Recon] IDEAL-IQ Echo Phase Correction Robustness Improvement
HCSDM00713353	[MR29.1_R04] [PSD]HCSDM00: HAST:data acquisition error with A193 3D SAG VIBRANT Unilateral base ASPIR ASSET no turbo protocols
HCSDM00713444	[MR29.1_R04] [PSD] MPRAGE PROMO download failed with minimum Recovery Time
HCSDM00713448	[MR29.1_R04] [PSD] Dual Echo Artifact when cv6 is on
HCSDM00713602	[MR29.1_R04] [PSD]Enable SAR optimization (UserCV29) for 1.5T first and normal mode
HCSDM00713608	[MR29.1_R04] [host] Prone Position Axial Breast Image Rotate Feature
HCSDM00713706	[MR29.1_R04] [Mulan/HD] fgre-xl 3plane not able to be saved
HCSDM00713720	[MR29.1_R04] Fast multiNEX Calibration
HCSDM00713725	[MR29.1_R04] Include pfv control in PUREMIX
HCSDM00713743	[MR29.1_R04] [PSD]SSFSE with TRF protocol crash with calcLopeVarflip error
HCSDM00713839	[MR29.1_R04] [Systo]Update fixed R1/R2 value in signal acquisition for 1.5T system with MEGASWITCH
HCSDM00714190	[MR29.1_R04] [subsystem] Promote GE 8ch Foot Ankle and Invivo 8ch Foot Ankle coil to open ARC by customers' request
HCSDM00714191	[MR29.1_R04] subsyste update calibration Dir. in prescan for Invivo 8ch Wrist and 8ch Foot ankle coil, GE 8ch Foot ankle coil.
HCSDM00715219	[MR29.1_R04] [host] Auto transfer by series cannot transfer images after moving the cradle
HCSDM00715220	[MR29.1_R04] [host] Reformat sequence is automatically transferred without permission Issue
HCSDM00715638	[MR29.1_R04] [PSD]Partial NEX over one should not be supported for BRAVO MP-RAGE

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HCSDM00716723	[MR29.1_R04] [Systools] Back EMF and coolwave promote for HRMW testing
HCSDM00716874	[MR29.1_R04] [host] ARDL DW-EPI runs without ARDLdwepi option key
HCSDM00717085	[MR29.1_R04] [PSD]DWI Propeller download failed with Acoustic Reduction (ART) imaging option
HCSDM00717088	[MR29.1_R04] update Default Freq. direction &Calibration Freq. direction for 1.5T coils.
HCSDM00717095	[MR29.1_R04] [PSD] ghost artifacts in multi-slice Ax ungated Fiesta
HCSDM00717107	Acquisition failure in slice ASSET with R=3
HCSDM00717111	[MR29.1_R04] [recon] Flex long recon lag due to incorrect zFOV calculation
HCSDM00717787	[MR29.1_R04] [host]Calibration coverage logic to determine if another cal is required is broken.
HCSDM00717811	[MR29.1_R04] [IPM] HCSDM00717811- Adding few extra parameters to MR scanner metadata being sent to IPM Cloud
HCSDM00717901	[MR29.1_R04] [host]Shoulder image dark line artifact from Atlantis pilot site
HCSDM00718008	[MR29.1_R04] [Apps] Silenz MRA architect GE protocol has wrong HNU coil
HCSDM00719433	[MR29.1_R04] [swtools] update R04 mapping to support OM installation
HCSDM00719799	Integrate Option Key MR30. to MR29.1 build
HCSDM00720270	[MR29.1_R04] [host]Series test page annotation for imaging calibration is incorrect for prescan options string.
HCSDM00720271	[MR29.1_R04] [host]Incorrect series reused for imaging calibration.
HCSDM00720505	[29.1_R04] [swtools] Create an option key ARDLdwepi
HCSDM00720680	[MR29.1_R04] [swtools]VX29.1_R02 (and RX29.1) Root Partition Full issue
HCSDM00720969	[MR29.1] [Recon]Prescan time impacted by oversampled scan archive compression with performance ICE
HCSDM00720970	Add logs to track memory freeing of pointer variable <grx3PlnData> causing a host shut down due to corruption/double free.
HCSDM00721124	[MR29.1_R04] [host] i18n support for Prone Position Axial Breast Image Rotate Feature
HCSDM00723404	[MR29.1_R04] Add 8ch TR Knee coil by RFT A Port for Mulan
HCSDM00724241	[MR29.1_R04] [recon] Coronal LAVA-Flex Robustness Improvement
HCSDM00725856	"Interleave" still available with type-in for spacing on MUSE DWI.
HCSDM00726046	CSO # 00826201, Haystacks Imaging LLC, SID # 301723MR450W, 1.5T Optima MR450w GEM
HCSDM00726518	[MR29.1_R04] [host] Performance degradation Issue with more number of slices in viewEdit/Setup Workflow.
HCSDM00722726	[MR29.1_R04] [host] Breast Image Orientation based on Option Key
HCSDM00728572	System cannot detect HEC via cable p/n 5729973 changed under SCR-0131764